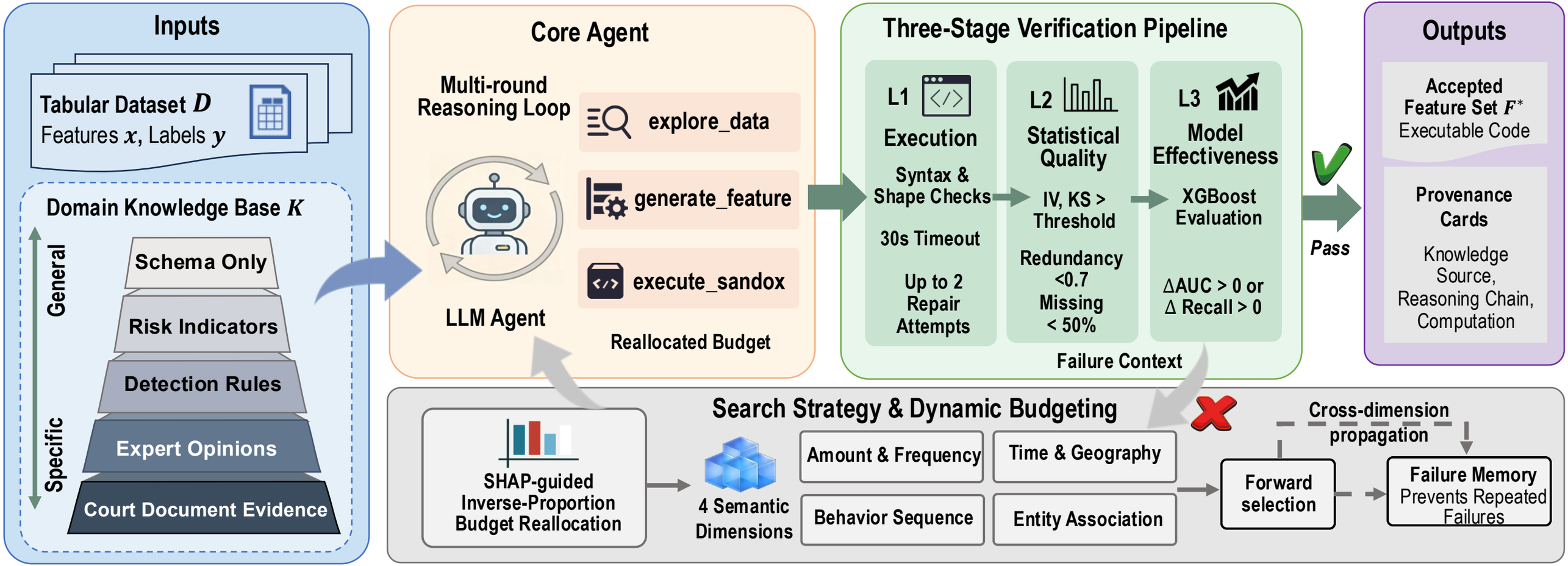

Inputs
Tabular Dataset D
Features x, Labels y
Domain Knowledge Base K
General
Specific
Schema Only
Risk Indicators
Detection Rules
Expert Opinions
Court Document Evidence
Core Agent
Multi-round Reasoning Loop
explore_data
generate_feature
execute_sandox
LLM Agent
Reallocated Budget
Three-Stage Verification Pipeline
L1
Execution
Syntax & Shape Checks
30s Timeout
Up to 2 Repair Attempts
L2
Statistical Quality
IV, KS > Threshold
Redundancy <0.7
Missing < 50%
L3
Model Effectiveness
XGBoost Evaluation
ΔAUC > 0 or Δ Recall > 0
Pass
Failure Context
Outputs
Accepted Feature Set F^*
Executable Code
Provenance Cards
Knowledge Source, Reasoning Chain, Computation
Search Strategy & Dynamic Budgeting
SHAP-guided Inverse-Proportion Budget Reallocation
4 Semantic Dimensions
Amount & Frequency
Time & Geography
Behavior Sequence
Entity Association
Cross-dimension propagation
Forward selection
Failure Memory
Prevents Repeated Failures